

A Single Chain Suffices: Re-examining Diversity-Enhanced flowMC Ensembles under Equal Sampling Budgets

Mingyu Shi¹

¹Independent Researcher

Abstract

Normalizing-flow Markov chain Monte Carlo (MCMC), of which flowMC [1,2] is a representative implementation, adds a learned global proposal on top of local moves so that difficult target densities become easier to sample. A natural idea for making such a sampler more reliable is to run several of them with different architectures and pool their samples. This paper began as an attempt to validate exactly that: a five-member diverse flowMC ensemble equipped with three aggregation mechanisms — ESS-based adaptive tempering, coverage-aware reweighting, and quality-based member weighting. On six benchmark targets in 20 and 50 dimensions, the ensemble seemed to reduce the average marginal Jensen–Shannon (JS) distance by about 18% relative to a single flowMC chain, significantly on every configuration. One control reversed the story: the ensemble draws five times as many samples as the baseline (10,000 against 2,000), and histogram-based JS distance carries a finite-sample bias. At a matched 10,000-sample budget, a single chain beats the ensemble on nine of eleven configurations, ties on the other two, lowers the mean JS by a further 24%, and runs about six times faster. We show that the JS estimate of a perfect sampler follows the closed form $\sqrt{c(1/N + 1/M)}$ with $c \approx (B_{\text{eff}} - 1)/8$ nats, which explains the original “gain” almost entirely. We then reproduce the ensemble’s deficit *without running flowMC at all*: applying the released aggregation operators to exact draws from the targets recovers about 90% of the gap, and adding the tempering schedule reproduces the measured ensemble values on the correlated and funnel targets. The damage decomposes into a genuine distribution shift caused by the coverage and tempering reweighting — confirmed by an unbiased energy distance — and a pure estimator-level loss caused by resampling with replacement, which halves the effective histogram sample. Uniform pooling is harmless and matches the single chain, and adding members (up to twenty) does not repair the aggregation. An audit of the reference generators additionally shows that two targets were scored against a law that differs from their potential, reframing which targets are genuinely sampler-limited. The recommendation is unchanged and now mechanistic: run one flowMC chain at the full budget; if members are pooled, pool them uniformly; and avoid reweight-and-resample aggregation.

Keywords: Markov chain Monte Carlo; normalizing flows; ensemble sampling; negative results; Jensen–Shannon divergence; finite-sample bias; divergence estimation

1 Introduction

Sampling from complex, high-dimensional and multimodal distributions is one of the recurring difficulties in Bayesian inference [5]. Normalizing-flow MCMC, such as flowMC [1,2], combines local gradient-based moves with a global proposal drawn from a normalizing flow [3,4] that is trained

on the running samples. On targets where purely local samplers move slowly, this kind of method usually does well.

When we began this work, our reasoning was the common one: if a single flow-based sampler is good, then a group of them, built with different flow architectures and different step budgets, should be even more reliable, because the weak points of one member could be covered by the others. Ensemble and population MCMC have a long tradition along this line [6,7,8]. So we designed a diverse five-member flowMC ensemble and added three lightweight aggregation mechanisms, each aimed at a failure mode we expected: adaptive tempering for members that explore poorly, coverage-aware reweighting for regions that get missed, and quality-based member weighting so that weaker members count less.

At first the results agreed with our expectation. Across six targets in 20 and 50 dimensions, the ensemble lowered the average marginal JS distance to a reference sample by about 18% relative to a single chain, and on every configuration the difference was statistically significant. We were ready to write this up as a positive method.

The reason this paper is a negative result instead is one control experiment. With the budget allocation we used, called equal-per-member, every member draws as many samples as the single-chain baseline, so five members produce five times the samples in total — 10,000 against 2,000. Histogram-based JS distance is not a fair quantity to compare across different sample sizes, because it carries an upward bias that shrinks as the number of samples grows. The honest comparison is therefore not a single chain at 2,000 samples, but a single chain at the same total of 10,000.

We ran it, and it reversed the conclusion. A single flowMC chain at 10,000 samples is better than the ensemble on nine of eleven configurations, ties on the two remaining ones, and lowers the mean JS by a further 24%. The ensemble never wins. We then checked the metric itself by drawing directly from the true distributions, and found that even a perfect sampler’s marginal JS estimate moves from about 0.085 at 2,000 samples to about 0.043 at 10,000 — a factor close to two, and almost the same number for every target. A second-order analysis of the estimator (Section 5.3) shows why: the squared JS estimate of a perfect sampler is linear in $1/N + 1/M$ with a slope set by the number of histogram bins, so the floor is a property of the metric, not of the target. This accounts for the original “gain” on its own.

After that, we kept digging, and the ensemble came apart piece by piece. A homogeneous ensemble of five identical members performs the same as the diverse one, so diversity does nothing here. None of the three aggregation mechanisms helps in the ablation; two of them slightly hurt. When we throw away the aggregation step entirely and simply concatenate the members’ raw samples, the accuracy returns to the single-chain level. Finally — and this is the part that turns the negative result into an explanation — we removed flowMC itself: we applied the released aggregation operators to *exact* draws from the targets, and the deficit reappeared almost unchanged. About 90% of the gap between the ensemble and the matched single chain is reproduced by the aggregation acting on perfect samples, and adding the adaptive-tempering schedule reproduces the measured ensemble values on the correlated and funnel targets to within a few thousandths. The mechanism splits cleanly in two: the coverage and tempering reweighting genuinely distorts the sampled distribution (an unbiased energy distance confirms the shift), while the resampling-with-replacement step leaves the distribution intact but duplicates points, which halves the effective sample behind the histogram and pushes the biased JS estimator upward. Pooling the members is harmless; every part of the machinery we put around the pooling either distorts the law or degrades the estimate.

We think this is worth reporting, precisely because the positive version of the story is the easy one to believe and the easy one to publish. The single experiment that breaks it is cheap. The contributions of this paper are:

1. We show that the apparent advantage of a diversity-enhanced flowMC ensemble over a single chain is, for these targets and budgets, an artifact of comparing at unequal sample counts, and we give the closed-form finite-sample law of the metric, $\mathbb{E}[\widehat{\text{JS}}^2] \approx c(1/N + 1/M)$ with $c \approx (B_{\text{eff}} - 1)/8$ nats, that quantifies the artifact and is verified across sample sizes and bin counts.
2. With a matched-budget control, a single chain matches or beats the ensemble on every configuration (better on nine of eleven, tied on the two banana cases) and is also about six times faster; plain uniform pooling of the members recovers single-chain accuracy, which localizes the entire deficit in the aggregation step.
3. We reproduce and decompose the deficit *without flowMC*: the released aggregation operators applied to exact draws recover about 90% of the gap, and with the tempering schedule they reproduce the measured ensemble on the correlated and funnel targets. The decomposition separates a genuine distribution shift (coverage and tempering reweighting; confirmed under an unbiased energy distance) from a pure estimator-level loss (resampling with replacement, which doubles the sampler-side variance term exactly as the bootstrap predicts). Increasing the ensemble size to twenty members does not repair the aggregation.
4. We document two practical methodological warnings: histogram-based divergences are only comparable at matched sample counts and should be reported together with their finite-sample floor; and reference generators must be validated against the potential — an audit shows that for two of our six targets the released reference law differs from the law defined by the potential, which changes the interpretation of which targets are sampler-limited.

2 Background and Related Work

2.1 Normalizing-flow MCMC

A normalizing flow writes a distribution as an invertible, learnable transformation of a simple base density, with a tractable Jacobian [3,4]. flowMC [1,2] uses such a flow as a global independence proposal together with local Metropolis-adjusted Langevin steps, and trains a rational-quadratic neural spline flow [4] in alternating training and production phases. In this work we keep flowMC unchanged both for the single-chain baseline and for each ensemble member, so that any difference we see comes from ensembling and aggregation, not from a modified kernel.

2.2 Ensemble and population MCMC

Combining several chains to gain reliability is an old idea, including affine-invariant ensemble samplers [6] and their popular implementation [7], as well as population-based and tempering methods more generally [8,9]. Our setup is different in one respect: the members do not talk to each other while sampling. They are independent flowMC runs and are combined only at the aggregation stage. This isolates the effect of pooling several members from any inter-chain communication.

2.3 Evaluating sample quality

We measure the marginal JS distance [11] between the sampler output and a large reference sample, averaged over the dimensions, together with effective sample size [12] and runtime. A point that becomes central later (Section 5.3) is that histogram-based divergence estimators carry a positive

finite-sample bias of the same origin as the classical bias of plug-in entropy estimates [14,15], so a comparison made at different sample counts can mislead. We also use the energy distance [16], whose unbiased U-statistic form has expectation zero when the two samples follow the same law, as a joint-distribution check that is free of this bias; kernel two-sample statistics such as MMD [17] would serve the same purpose.

3 Methods

3.1 Single flowMC baseline

The baseline is one flowMC sampler with two training and two production loops, ten local MALA steps and ten global flow steps per loop, and five training epochs per loop. The flow is a rational-quadratic spline with three coupling layers, eight bins, and two hidden layers of width 64. These settings are kept fixed across all targets. We run this baseline at two sample sizes: 2,000 particles, which was our original baseline, and 10,000 particles, which is the matched control whose total equals the ensemble. We call this single flowMC run the “single-chain” baseline, following MCMC terminology, although each run returns a set of particles rather than one sequential chain.

3.2 Diverse ensemble

The ensemble has five flowMC members that differ in capacity and step allocation: hidden-layer widths $((64, 64), (96, 96), (128, 64), (64, 128), (32, 64, 32))$, number of spline layers (three or four), number of bins (eight to twelve), and the split between local and global steps. The members are independent and are combined only at the aggregation step.

3.3 Aggregation mechanisms

Three mechanisms act at the aggregation stage. **Adaptive tempering** changes a per-member temperature T in $\log \pi_T(x) = \frac{1}{T} \log \pi(x)$: after each member, T is multiplied by 1.1 (capped at 2) if that member’s ESS ratio falls below a target of 0.5, and by 0.95 (floored at 0.1) otherwise, starting from $T = 1$. **Coverage-aware reweighting** clusters the pooled samples on their first two coordinates with k -means ($K = 8$, five initializations) and gives inverse-frequency weights $w_i \propto 1/\text{count}(c_i)$, followed by a rare-cluster boost $w \leftarrow w(1 + 0.35(w/\max w - 1))$. **Quality-based member weighting** scores members by

$$s_m = \text{ESS}_m^\beta \exp(-\gamma \text{JS}_m) \exp(-\lambda \sigma_{\log p, m}), \quad (1)$$

with $\beta = 1$, $\gamma = 2$ and $\lambda = 0.05$, standardizes the scores, and turns them into weights with a softmax at temperature $\alpha = 0.35$. Each mechanism was chosen because it targeted a failure mode we believed mattered. As Sections 5.4 and 5.6 report, none of them earns its place.

3.4 Pooling and budget allocation

The pooled samples are combined by multiplying the member weights and the coverage weights, normalizing, and **resampling with replacement** to a target count of (per-member size) $\times$ (number of members). Under **equal-per-member**, each member draws 2,000 samples, so the ensemble total is 10,000. Under **equal-total**, the 2,000-sample budget is split across members (400 each). The equal-per-member ensemble therefore uses five times the compute and produces five times the samples of the 2,000-particle single chain, and the fair comparison is the 10,000-particle single chain.

4 Experimental Setup

4.1 Targets and protocol

We use six targets that stress different weaknesses of a sampler: a banana density (strong nonlinear correlation), a bimodal Gaussian (separated modes), Neal’s funnel [10] (varying scale), a correlated Gaussian (correlation 0.95), the Rosenbrock density (curved valley), and a Gaussian ring (eight modes on a ring). All are run in 20 and 50 dimensions except the ring, which is 20D only, giving eleven configurations. Each one is repeated five times with deterministic seeds, the single-chain and ensemble runs in a repetition share a seed stream, and the reference samples (20,000 per target) are drawn from the generative form of each target. Section 5.7 audits these reference generators against the potentials. All seeds, configurations and code are released [13].

4.2 Metrics

The main metric is the average marginal JS distance computed on 100-bin histograms whose range per dimension is the 0.5–99.5 percentile span of the two samples, padded by 10%. We also report ESS from the integrated autocorrelation and the wall-clock runtime. The ESS we report is computed on the returned particle or pooled sample array and serves only as a diagnostic of duplication and correlation in the returned sample set; it should not be read as a full trajectory-based MCMC mixing ESS. Where useful we give paired t -tests and Cohen’s d over the five repetitions; tables report means $\pm$ one standard deviation over the repetitions.

5 Results

5.1 The apparent ensemble gain

Against the 2,000-particle single chain, the equal-per-member ensemble lowers the marginal JS distance on all eleven configurations, by 6.8% to 29.2% (mean 18.2%), and every configuration is significant at $p < 0.05$. Read on its own, this looks like a clear success for ensembling, and the largest improvements are on the multimodal and structured targets. The rest of this section explains why it is not what it looks like.

5.2 The equal-budget control

The equal-per-member ensemble draws 10,000 samples in total, while the baseline it was compared with draws only 2,000. The fair comparison is a single chain at 10,000 samples, which we give in Table 1 and Figure 1. The single 10,000-particle chain reaches a mean JS of 0.065, against 0.085 for the ensemble and 0.104 for the 2,000-particle baseline. It beats the ensemble on nine of eleven configurations, ties on the two banana cases (within repetition noise), with 17% to 44% improvements on exactly those targets where the ensemble had looked strongest (bimodal, funnel, correlated, ring).

The single-chain advantage is not marginal where it exists: the per-configuration gap is several to over a hundred times the single-chain repetition standard deviation. The two banana cases are the exception, and they are informative rather than awkward. There, both the single chain and the ensemble sit far above the noise floor (JS around 0.10–0.14 against a floor of 0.043), which tells us that flowMC is simply failing to capture the banana’s strong nonlinear correlation, no matter the sample count or the ensembling. This is a sampler-limited regime, not a sample-limited one, and the near-tie reflects that neither extra samples nor extra members help. So the banana is not a case

where ensembling succeeds; it is a case where nothing we tried does. (The Rosenbrock and ring residuals above the floor turn out to have a different explanation; see Section 5.7.)

Table 1: Marginal JS distance (mean $\pm$ sd over five repetitions). The equal-budget single chain on the right is the proper control for the ensemble in the middle. Lower is better. “Tie” marks differences within repetition noise.

Target	Dim	Single @2,000	Ensemble @10,000	Single @10,000	Single@10k vs ens
Banana	20	0.169 $\pm$ 0.005	0.138 $\pm$ 0.004	0.137 $\pm$ 0.005	tie
Banana	50	0.127 $\pm$ 0.003	0.095 $\pm$ 0.003	0.097 $\pm$ 0.007	tie
Bimodal	20	0.087 $\pm$ 0.002	0.061 $\pm$ 0.001	0.045 $\pm$ 0.002	−27%
Bimodal	50	0.086 $\pm$ 0.001	0.061 $\pm$ 0.000	0.045 $\pm$ 0.001	−27%
Correlated	20	0.087 $\pm$ 0.001	0.081 $\pm$ 0.002	0.045 $\pm$ 0.001	−44%
Correlated	50	0.086 $\pm$ 0.001	0.071 $\pm$ 0.001	0.044 $\pm$ 0.000	−38%
Funnel	20	0.086 $\pm$ 0.002	0.074 $\pm$ 0.002	0.044 $\pm$ 0.000	−40%
Funnel	50	0.087 $\pm$ 0.001	0.079 $\pm$ 0.001	0.045 $\pm$ 0.000	−43%
Ring	20	0.097 $\pm$ 0.001	0.071 $\pm$ 0.001	0.059 $\pm$ 0.002	−17%
Rosenbrock	20	0.129 $\pm$ 0.001	0.114 $\pm$ 0.003	0.090 $\pm$ 0.001	−20%
Rosenbrock	50	0.103 $\pm$ 0.001	0.089 $\pm$ 0.001	0.062 $\pm$ 0.000	−30%
Mean		0.104	0.085	0.065	−24%

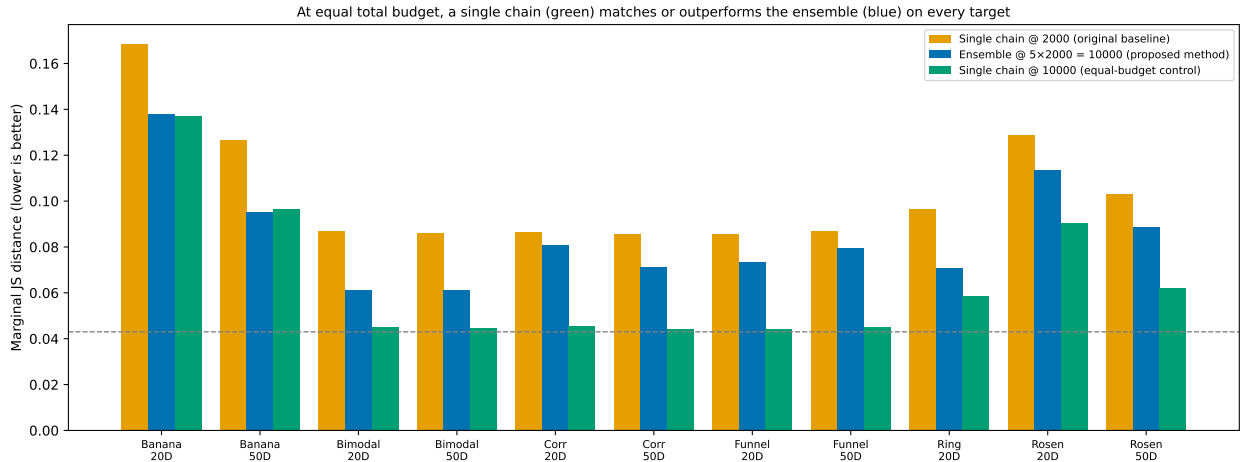


Figure 1: At equal total budget, the single 10,000-particle chain (green) matches or outperforms the ensemble (blue) on every target, winning on nine of eleven configurations and tying on the two banana cases. The ensemble’s apparent advantage over the 2,000-particle baseline (orange) does not survive the matched comparison. The dashed line is the finite-sample noise floor of a perfect sampler at 10,000 samples (Section 5.3).

5.3 Why: marginal JS distance is saturated by finite-sample bias

The reason behind the reversal is simple, and we should have checked it earlier. Histogram-based JS distance is a biased estimator: it reports a larger divergence when the sample is small, and the bias goes down as the sample grows. Drawing directly from the true distributions, we find that the marginal JS estimate of a perfect sampler is remarkably constant across all eleven configurations:

0.085–0.086 at 2,000 samples and 0.043–0.044 at 10,000, a factor close to two regardless of the target or the dimension. The floor is set by the number of histogram bins and the sample size, not by the target.

This behaviour has a simple closed form. Write $\hat{p}$ and $\hat{q}$ for the per-dimension histograms of the sampler output (N points) and the reference ($M = 20,000$ points), both drawn from the same law. A second-order (χ^2) expansion of the plug-in JS divergence around $p = q$, of the same kind that gives the classical bias of plug-in entropy estimates [14,15], yields

$$\mathbb{E}[\widehat{\text{JSD}}] \approx \frac{1}{8} \sum_b \frac{\mathbb{E}[(\hat{p}_b - \hat{q}_b)^2]}{m_b} \approx c \left(\frac{1}{N} + \frac{1}{M} \right), \quad c \approx \frac{B_{\text{eff}} - 1}{8} \text{ nats}, \quad (2)$$

so the JS *distance* floor is $\sqrt{c(1/N + 1/M)}$, where B_{eff} counts the effectively occupied bins and absorbs the sparse-bin corrections to the χ^2 approximation. The data follow this law closely (Figure 2): fitting c on the measured floor curve gives $c = 13.4$ nats, i.e. $B_{\text{eff}} \approx 108$ for $B = 100$ bins; the squared floor is linear in $1/N + 1/M$ at every bin count we tried, with fitted slopes $c = 3.2, 6.5, 13.1$ and 26.5 for $B = 25, 50, 100$ and 200 , i.e. $c \approx 0.13 B$; and the parameter-free prediction for the ratio of the two floors, $\sqrt{(1/2000 + 1/M)/(1/10000 + 1/M)} = 1.91$, matches the measured factor of about 1.95–2.0 to within a few per cent. Equation (2) also explains why the floor is the same for every target: to leading order it does not depend on the underlying density at all, only on the binning and the two sample sizes.

The 2,000-particle single chain sits at about 0.104 on average; on the Gaussian-like targets it is already at its 2,000-sample floor (around 0.086), with the mean pulled up by the banana and Rosenbrock cases (Section 5.7 examines how much of the latter is sampler error). So on the targets where the metric can resolve sampler quality, the single chain already reproduces the marginals nearly as well as the metric can see at that sample size, and the original 18% “gain” of the ensemble is for the most part just the difference between the 2,000-sample and 10,000-sample floors.

Figure 2(a) makes this concrete and also previews the ensemble’s problem. The single chain at 2,000 samples sits just above the 2,000-sample floor, and the single chain at 10,000 sits just above the 10,000-sample floor. Both of them are near-optimal for the metric at their sample size. The full ensemble, although it nominally produces 10,000 samples and its measured effective sample size is close to that (about 9,800 by integrated autocorrelation), still lands at a JS value (0.085) that a single chain reaches with only about 2,000 samples. So the ensemble is not short of effective samples; something in the way they are aggregated keeps the histogram from improving the way more samples should. We return to this in Sections 5.5 and 5.6.

Two things follow. First, the comparison in Section 5.1 was not informative, because it placed the two samplers at different points on the bias curve. Second, marginal JS distance is a weak way to tell samplers apart in this regime, and any benchmark using it should either match the sample counts exactly or report the floor of Equation (2) alongside the results. We learned this the hard way.

5.4 Ablation: the aggregation mechanisms do not help

We turn off each mechanism in turn (equal-per-member) and compare to the full ensemble. None of them gives a net benefit (Table 2, Figure 3). Removing coverage-aware reweighting improves the mean JS by 5.4% (best on the correlated Gaussian, where it improves JS by 22% in 20D); removing adaptive tempering improves it by 4.3%; removing member weighting changes it by under 1%. The full method is the worst of the four ensemble variants. A homogeneous ensemble of five identical members performs essentially the same as the diverse one (0.0840 versus 0.0849, a difference

The marginal-JS noise floor of a perfect sampler is explained by finite-sample estimator bias

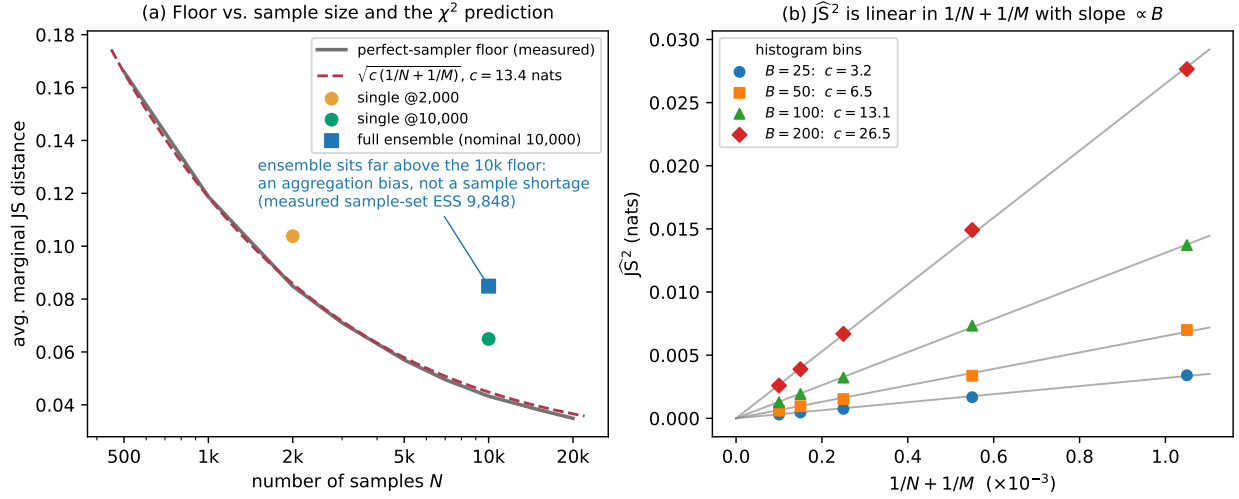


Figure 2: (a) The marginal JS noise floor of a perfect sampler falls with the number of samples (grey curve, averaged over the correlated, bimodal and funnel targets; the same dependence holds across all eleven configurations) and is matched by the closed form $\sqrt{c(1/N + 1/M)}$ of Equation (2) (dashed). The single chains at 2,000 and 10,000 samples sit close to their respective floors, while the full ensemble, plotted at its nominal 10,000 samples, sits far above the 10,000-sample floor even though its measured sample-set ESS is near 10,000: the gap is a bias introduced by the aggregation rather than a shortage of samples. (b) The squared floor is linear in $1/N + 1/M$ with slope proportional to the number of histogram bins B , as Equation (2) predicts; points are measured, lines are least-squares fits through the origin.

inside the repetition noise), so member diversity is not the active ingredient either. This was more negative than we expected when we designed the mechanisms, and together with Section 5.2 it says clearly that neither diversity nor the aggregation machinery earns its place. These are one-at-a-time ablations rather than a full factorial over all mechanism combinations; Section 5.6 complements them with a constructive decomposition that adds the mechanisms one by one to exact samples.

Table 2: Ablation under the equal-per-member budget: marginal JS distance averaged over the eleven configurations (mean $\pm$ sd of the configuration-averaged JS over the five repetitions). Lower is better. All ensemble variants, including the homogeneous one, are still beaten by the single 10,000-particle chain.

Variant	Mean JS	vs. single @2,000	vs. full ensemble
Single chain @2,000	0.1038 ± 0.0008	—	+22.2%
Full ensemble	0.0849 ± 0.0002	−18.2%	—
No coverage reweighting	0.0803 ± 0.0005	−22.6%	−5.4%
No adaptive tempering	0.0813 ± 0.0003	−21.7%	−4.3%
No member weighting	0.0842 ± 0.0004	−18.9%	−0.8%
Homogeneous ensemble (5 $\times$ identical)	0.0840 ± 0.0001	−19.1%	−1.1%
Single chain @10,000	0.0649 ± 0.0009	−37.5%	−23.6%

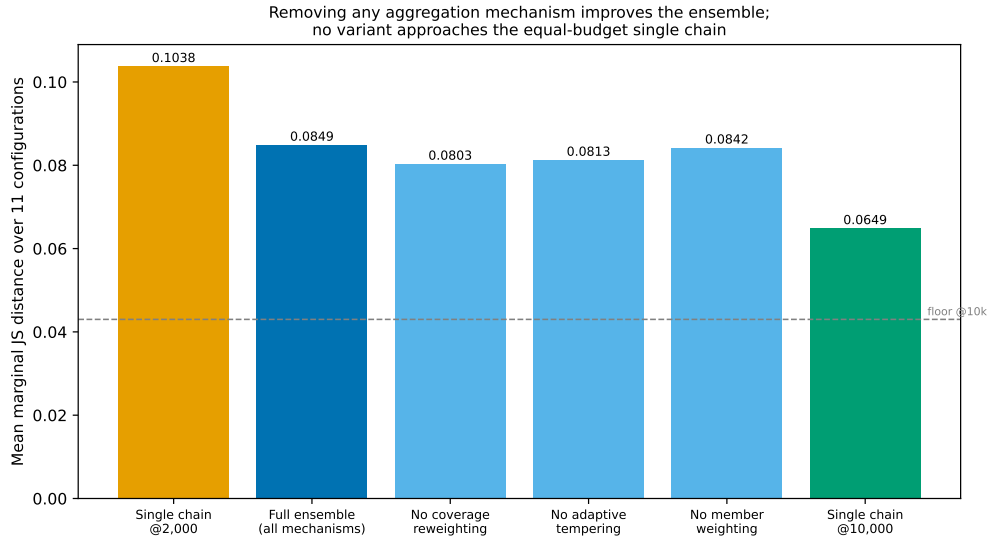


Figure 3: One-at-a-time ablation under the equal-per-member budget (mean marginal JS over the eleven configurations). Removing any of the three mechanisms improves the ensemble, so the full method is the worst of the four variants, and no ensemble variant approaches the equal-budget single chain. The dashed line is the 10,000-sample noise floor.

5.5 Localizing the cause: uniform pooling versus the aggregation machinery

The matched-budget control tells us the full ensemble loses, but it does not tell us which part of the ensemble is responsible. To find out, we replace the aggregation (the weighted resampling-with-replacement and the coverage, tempering and weighting mechanisms) with the simplest possible alternative: we just concatenate the five members’ raw 2,000-sample outputs into one pool of 10,000

unique samples, with no resampling and no weights. Across all eleven configurations, this uniform pool matches the single 10,000-particle chain (mean JS 0.062 against 0.065) and sits far below the full ensemble (0.085); it matches the single chain on every target within repetition noise, and is slightly better on the two hardest ones, the banana and the ring, where pooling several independent runs seems to help a little (Table 3, Figure 4). We do not over-read this small edge, since we did not separate it from a plain multi-start effect, but the main point is clear: pooling does not hurt.

Table 3: Marginal JS distance (mean $\pm$ sd over five repetitions) across all eleven configurations. Plain uniform pooling of the members recovers single-chain accuracy; the full aggregation machinery does not. Lower is better.

Target	Dim	Full ensemble	Uniform pool	Single @10,000
Banana	20	0.138 $\pm$ 0.004	0.128 $\pm$ 0.002	0.137 $\pm$ 0.005
Banana	50	0.095 $\pm$ 0.003	0.084 $\pm$ 0.001	0.097 $\pm$ 0.007
Bimodal	20	0.061 $\pm$ 0.001	0.044 $\pm$ 0.001	0.045 $\pm$ 0.002
Bimodal	50	0.061 $\pm$ 0.000	0.044 $\pm$ 0.000	0.045 $\pm$ 0.001
Correlated	20	0.081 $\pm$ 0.002	0.044 $\pm$ 0.001	0.045 $\pm$ 0.001
Correlated	50	0.071 $\pm$ 0.001	0.044 $\pm$ 0.000	0.044 $\pm$ 0.000
Funnel	20	0.074 $\pm$ 0.002	0.044 $\pm$ 0.000	0.044 $\pm$ 0.000
Funnel	50	0.079 $\pm$ 0.001	0.045 $\pm$ 0.000	0.045 $\pm$ 0.000
Ring	20	0.071 $\pm$ 0.001	0.055 $\pm$ 0.001	0.059 $\pm$ 0.002
Rosenbrock	20	0.114 $\pm$ 0.003	0.090 $\pm$ 0.000	0.090 $\pm$ 0.001
Rosenbrock	50	0.089 $\pm$ 0.001	0.062 $\pm$ 0.000	0.062 $\pm$ 0.000
Mean		0.0849	0.0622	0.0649

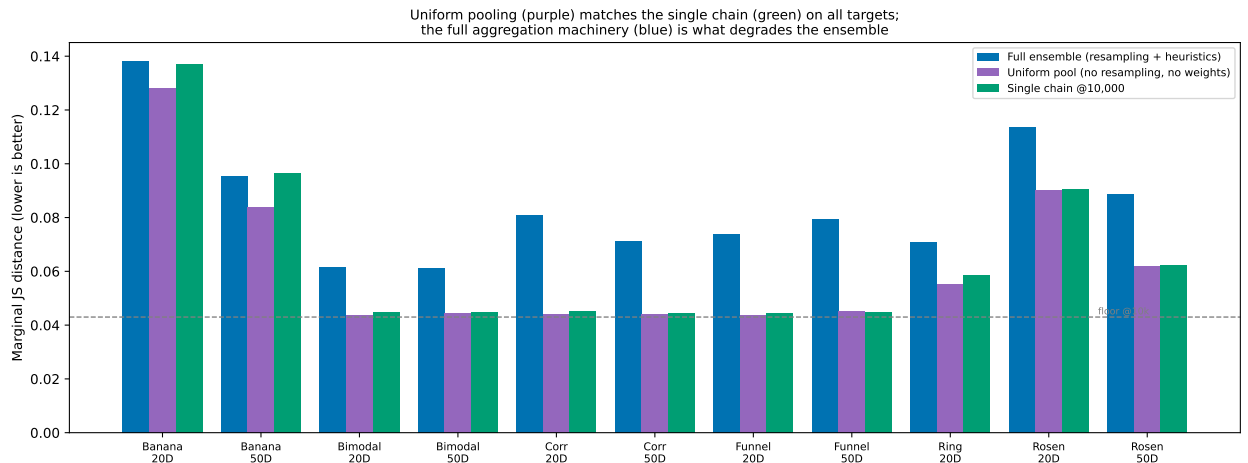


Figure 4: Across all eleven configurations, a plain uniform pool of the five members’ raw samples (purple) reaches the same accuracy as a single chain at the matched budget (green), while the full aggregation machinery (blue) stays far worse. Pooling is harmless; the machinery around it is not.

So pooling the members is harmless on its own: a uniform pool of the same five members reaches the same accuracy as a single chain at the same budget. The whole deficit of the full ensemble comes from the aggregation machinery, not from the act of pooling. The mechanism is bias, not lost samples. The full ensemble and the uniform pool have almost the same measured effective sample size (about 9,800 and 9,700 respectively, both near the nominal 10,000), yet their

accuracy is far apart, so the deficit cannot be a shortage of effective samples. What differs is that the aggregation reweights the pooled samples by coverage and member quality and resamples from those weights, which pulls the sample distribution away from the target and biases the marginals; the uniform pool instead preserves the empirical distribution produced by the member samplers without imposing additional aggregation weights. The ablation supports this directly: removing the coverage and tempering reweighting is what improves the JS (Section 5.4). From the logged flowMC summaries alone we could not separate how much of the bias comes from the resampling-with-replacement and how much from the reweighting; Section 5.6 provides exactly this separation by applying the operators to exact samples.

This control also closes a confound that we worried about. One might suspect that the single 10,000-particle chain wins only because its flow is trained on more data (10,000 samples) than each ensemble member (2,000). But in the uniform pool, every member is still trained on only 2,000 samples, exactly like the ensemble members, and the pool of these members already matches the single chain. So the matched single-chain accuracy appears to be driven mainly by the total returned sample count under this marginal-JS diagnostic, not by how much data any individual flow was trained on. And the objection that the ensemble just needs a better-implemented pooling step does not hold either: the simplest pooling we could write already matches the single chain, and no aggregation we tested beats it.

5.6 A sampler-free decomposition of the aggregation bias

If the aggregation step is the culprit, it should damage *any* samples, not just flowMC’s. We test this directly by removing the sampler from the experiment: each member is replaced by 2,000 *exact* draws from the target’s generative law (the same generators that produce the reference samples of Section 4.1), and the released aggregation operators are applied to these perfect members at their default hyperparameters (Section 3.3), using the identical JS estimator. Five members of 2,000 draws are pooled per configuration, five repetitions per configuration; the resampling rng varies with the repetition so that the error bars include resampling noise. Because the members are exact, any elevation of the JS above the 10,000-sample floor is attributable to the operators alone. Table 4 and Figure 5 give the result as a bias budget in which the mechanisms are added one at a time.

Table 4: Sampler-free bias budget: marginal JS distance when the aggregation operators are applied to exact draws (mean over the eleven configurations $\pm$ sd of that mean over five repetitions). The χ^2 column is the prediction of Equation (2), with the bootstrap factor 2 on the sampler-side term for the resampled variants. The last row repeats the measured flowMC ensemble from Table 1 for comparison; the exact-sample stack reproduces about 90% of its deficit over the floor.

Variant (exact members, $5 \times 2,000$)	Mean JS	χ^2 prediction
Exact single draw of 10,000	0.0435 ± 0.0002	0.045
Uniform pool of $5 \times 2,000$ exact draws	0.0435 ± 0.0002	0.045
+ resampling with replacement (uniform weights)	0.0563 ± 0.0003	0.058
+ member weighting + resampling	0.0574 ± 0.0005	—
+ coverage reweighting + resampling	0.0783 ± 0.0007	—
Full aggregation stack (member $\times$ coverage, resampled)	0.0806 ± 0.0010	—
flowMC ensemble, measured (Table 1)	0.0849 ± 0.0002	—

The budget reads as follows. An exact single draw of 10,000 and the uniform pool of five exact members both sit on the floor (0.0435), reproducing Section 5.5 in idealized form: pooling

Sampler-free decomposition: the aggregation operators applied to exact samples reproduce ~90% of the ensemble's deficit

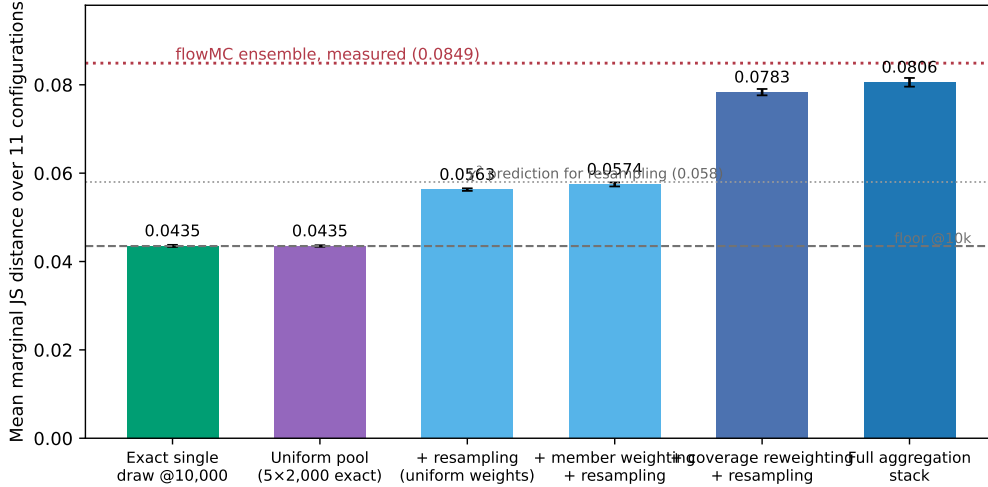


Figure 5: Sampler-free decomposition of the aggregation bias. Exact draws and their uniform pool sit on the 10,000-sample floor. Resampling with replacement alone lifts the JS to the level predicted by doubling the sampler-side variance term in Equation (2); member weighting adds almost nothing on top; coverage reweighting adds the largest distortion; and the full stack lands within 10% of the gap of the measured flowMC ensemble (dotted line) — without any sampler in the loop.

is exactly harmless. Resampling the pool with replacement at *uniform* weights — the step every aggregation variant shares — already lifts the JS to 0.0563. This is a pure estimator-level effect with a parameter-free prediction: bootstrap resampling doubles the variance of the empirical histogram [18], so Equation (2) puts the resampled pool at $\sqrt{c(2/N + 1/M)} = 0.058$, within a few per cent of what we measure. In other words, the resampling step throws away half of the effective sample behind the histogram while leaving the sampled law untouched. Member weighting adds almost nothing on top (0.0574): the five exact members are statistically identical, yet the standardize-then-softmax construction of Equation (1) still manufactures weights between 0.11 and 0.30 out of pure noise — a mildly cautionary detail, since a weighting scheme that cannot return to uniform on identical members will always inject some spurious imbalance. Coverage reweighting is the largest single contributor, lifting the mean to 0.0783, and the full stack reaches 0.0806 ± 0.0010 , which is 90% of the measured flowMC ensemble’s deficit over the floor (0.0849). The pooled sample-set ESS, meanwhile, is blind to all of this: 9,846 for the uniform pool and 9,848 for the fully aggregated pool, mirroring the values observed with real flowMC members and confirming that this diagnostic cannot detect the damage.

The size of the coverage distortion is geometry-dependent in an interpretable way. It is largest where k -means on the first two coordinates produces unbalanced clusters that the inverse-frequency weights then overcorrect — the banana (0.098 in 20D) and the Rosenbrock valley (0.123) — moderate on the correlated Gaussian and the funnel (0.078 and 0.070), and essentially absent on the bimodal Gaussian and the ring (0.058 and 0.056, i.e. the resampling-only level), whose symmetric mode structure yields balanced clusters and near-uniform weights. Consistent with a distortion concentrated on two of d coordinates, the coverage effect is diluted in 50D (mean 0.076 versus 0.081 in 20D).

Tempering. Adaptive tempering acts at the sampling stage rather than the aggregation stage, so the pool experiment above leaves it out. It can still be isolated exactly on the two target

families whose tempered laws $\pi^{1/T}$ are available in closed form. For healthy members the released schedule only ever takes the “sharpen” branch (ESS ratio ≥ 0.5), so the five members target $T = 1, 0.95, 0.9025, 0.857, 0.815$, i.e. powers $\pi^{1.00}$ up to $\pi^{1.23}$ — an increasingly *over-sharpened* target. For the correlated Gaussian, $\pi^{1/T}$ is the same Gaussian with covariance scaled by T ; for the funnel, the neck variable becomes $v \sim \mathcal{N}(\sigma_v^2(d-1)(T-1)/2, T\sigma_v^2)$ with $x_i | v \sim \mathcal{N}(0, Te^v)$, so the sharpening both shrinks and *shifts* the v marginal — by -0.41 at $d = 50$ and $T = 0.815$, more than one standard deviation of the original marginal ($\sigma_v = 0.3$). Pooling five exact draws from these tempered laws (no weights, no resampling) lifts the JS from the floor to 0.050 on the correlated Gaussian and to 0.064 (20D) and 0.084 (50D) on the funnel, so the schedule alone visibly biases perfect members, most strongly exactly where the paper’s ensemble looked worst. Applying the full aggregation stack to the tempered members then reproduces the measured flowMC ensemble values to within a few thousandths on all four configurations — 0.085 vs. 0.081 (correlated 20D), 0.076 vs. 0.071 (50D), 0.075 vs. 0.074 (funnel 20D) and 0.081 vs. 0.079 (50D) (Figure 6). On these targets, essentially the entire ensemble deficit is manufactured by the tempering and aggregation machinery operating on what are, at the marginal level, perfect samples.

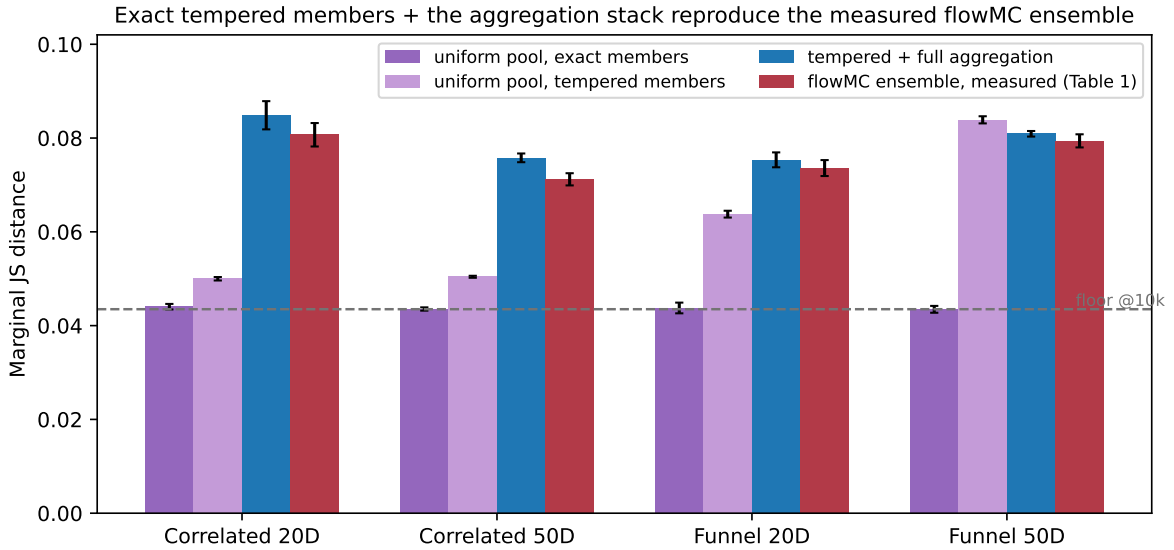


Figure 6: Exact isolation of the tempering schedule on the two target families whose tempered laws are available in closed form. A uniform pool of exact members sits on the floor; drawing the members from the schedule’s tempered targets $\pi^{1/T}$ (T down to 0.815) already biases the pool; and applying the full aggregation stack to the tempered members reproduces the measured flowMC ensemble (red) to within a few thousandths on all four configurations.

A joint metric separates the two mechanisms. The JS budget above mixes two conceptually different failures, and an unbiased joint metric cleanly separates them. Table 5 reports the energy distance [16] between each variant and the reference in the full 20D joint space, using the U-statistic estimator whose expectation is zero when the laws coincide. The exact draw, the uniform pool and — notably — the uniformly *resampled* pool are all at zero within noise ($|\text{ED}| \leq 0.002$ on every Table 5 entry, ≤ 0.009 for every individual repetition): resampling duplicates points but does not change the sampled law, so an unbiased metric does not see it. The full aggregation stack, in contrast, shows a genuine distribution shift exactly where the coverage weights are unbalanced — large on the

banana (0.56) and the Rosenbrock valley (7.05), small but nonzero on the correlated Gaussian and the funnel, and zero on the bimodal Gaussian and the ring. The deficit of the ensemble therefore has two distinct components: a *true* law distortion caused by the reweighting (and by tempering, which shifts the law by construction), visible to any consistent metric; and an *estimator-level* loss caused by the resampling, which only exists for finite-sample-biased metrics such as histogram divergences, but which any benchmark built on such metrics will faithfully punish. On the bimodal Gaussian and the ring, the full stack’s JS elevation is entirely of the second kind.

Table 5: Energy distance (unbiased U-statistic, 2,000-point subsamples, mean of five repetitions) between each exact-sample variant and the reference, 20D configurations. Zero means the laws coincide. Resampling leaves the law untouched; the full aggregation stack shifts it, most strongly where the coverage clusters are unbalanced. (Values are in the native scale of each target, so they are comparable within columns, not across them.)

Variant	Banana	Bimodal	Correlated	Funnel	Ring	Rosenbrock
Exact single draw of 10,000	0.000	-0.001	-0.000	0.000	-0.002	0.001
Uniform pool	0.000	-0.001	-0.001	0.000	0.001	0.001
Pool + uniform resampling	0.001	0.001	-0.000	0.001	0.000	0.001
Full aggregation stack	0.558	0.001	0.016	0.006	0.002	7.049

More members do not help. Finally, if the aggregation bias were a small-ensemble artifact, it should fade as members are added. It does not. Repeating the exact-sample experiment with $M = 2, 5, 10$ and 20 members of 2,000 draws each (Figure 7), the uniform pool tracks the exact-draw floor at every size, while the aggregated pool stays far above it at every size: at $M = 20$ (40,000 pooled samples) the aggregated pool only just reaches the accuracy of a plain 10,000-sample draw. Under this metric the aggregation effectively wastes about three quarters of the samples fed into it, and scaling the ensemble up cannot repair what the aggregation takes away.

5.7 An audit of the reference generators

The controlled experiments above also prompted us to audit the reference generators themselves — the “generative form of each target” of Section 4.1 — against the potentials that flowMC actually samples. For four of the six targets (banana, bimodal, funnel, correlated) the generator draws exactly from the law defined by the potential. For two it does not. The released Rosenbrock reference draws $x_0 \sim \mathcal{N}(a, 2^2)$ and $x_1 | x_0 \sim \mathcal{N}(x_0^2, 0.5^2)$, whereas the Rosenbrock potential $\frac{1}{2}[(a - x_0)^2 + b(x_1 - x_0^2)^2]$ with $b = 100$ implies $x_0 \sim \mathcal{N}(a, 1)$ and $x_1 | x_0 \sim \mathcal{N}(x_0^2, 0.1^2)$ — a reference that is substantially wider than the target in both coordinates. The released ring reference draws a uniform angle with radius $\mathcal{N}(5, 0.7^2)$ (a continuous annulus), whereas the potential is an eight-mode Gaussian mixture on the circle.

Table 6 quantifies the consequence: an exact 10,000-point draw from the *potential’s* law — the best any correct sampler of the potential can do — scores well above the floor against the released reference, entirely because of the first two coordinates. This changes the reading of two rows of Table 1. On the ring, the measured single chain at 10,000 (0.0585 ± 0.0018) is statistically indistinguishable from the perfect-sampler bound (0.0570): flowMC is essentially exact there, and the residual above the floor that we had implicitly read as sampler error is the reference mismatch. On the Rosenbrock, roughly half of the residual above the floor is mismatch (bound 0.069 in 20D against a measured 0.090; bound 0.053 in 50D against 0.062) and the remainder is genuine sampler

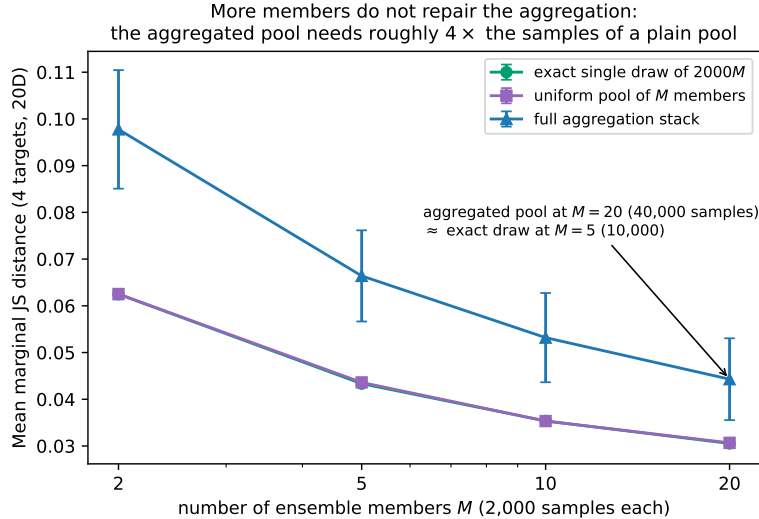


Figure 7: Ensemble-size scaling with exact members (mean over the bimodal, correlated, funnel and ring targets, 20D). The uniform pool of M members is indistinguishable from a single exact draw of the same total size, while the full aggregation stack needs roughly four times the samples to reach the same JS: at $M = 20$ it is only as good as an exact draw of $M = 5$.

error. The banana, whose generator is exact, remains the one target that is genuinely sampler-limited. None of the paper’s comparative conclusions changes — every method in every table is scored against the same reference, and the floors of Section 5.3 are reference-vs-reference quantities that do not depend on the potential at all — but the audit is a third methodological warning of the same family as the first two: a divergence-to-reference benchmark is only as meaningful as the reference generator, and the generator should be validated against the potential (for instance with exactly this exact-draw test, which is two lines of code once the generative form exists). Scripts that regenerate corrected references for both targets are included in the repository [13]; we keep Tables 1–3 under the original references for comparability with the released per-repetition data.

Table 6: Reference-generator audit. “Bound” is the marginal JS of an exact 10,000-point draw from the law defined by the *potential*, scored against the released 20,000-point reference (mean of five repetitions): the best value attainable by a perfect sampler of the potential. The mismatch is confined to the first two coordinates; the remaining dimensions sit on the floor. The measured single chain at 10,000 (Table 1) reaches the bound on the ring and approaches it on the Rosenbrock.

Target	Dim	Bound	JS dims 1–2	JS rest	Floor @10k	Single @10,000
Rosenbrock	20	0.0688	0.298	0.0433	0.0436	0.0904 ± 0.0009
Rosenbrock	50	0.0534	0.295	0.0434	0.0436	0.0621 ± 0.0003
Ring	20	0.0570	0.179	0.0435	0.0439	0.0585 ± 0.0018

5.8 Equal-total budget

A control on the other side splits the 2,000-sample budget across the five members (400 each), so the ensemble and the single chain produce the same total of samples. Here the ensemble is worse on all eleven configurations, by 11% to 64%, with large negative effect sizes. Members that are starved

of samples train weaker flows, and pooling them does not recover the quality of one chain that keeps the full budget. Together with Section 5.2 this brackets the conclusion from both directions: at an equal total of samples the ensemble loses whether the budget is small (2,000) or large (10,000). Figure 8 shows the per-configuration comparison.

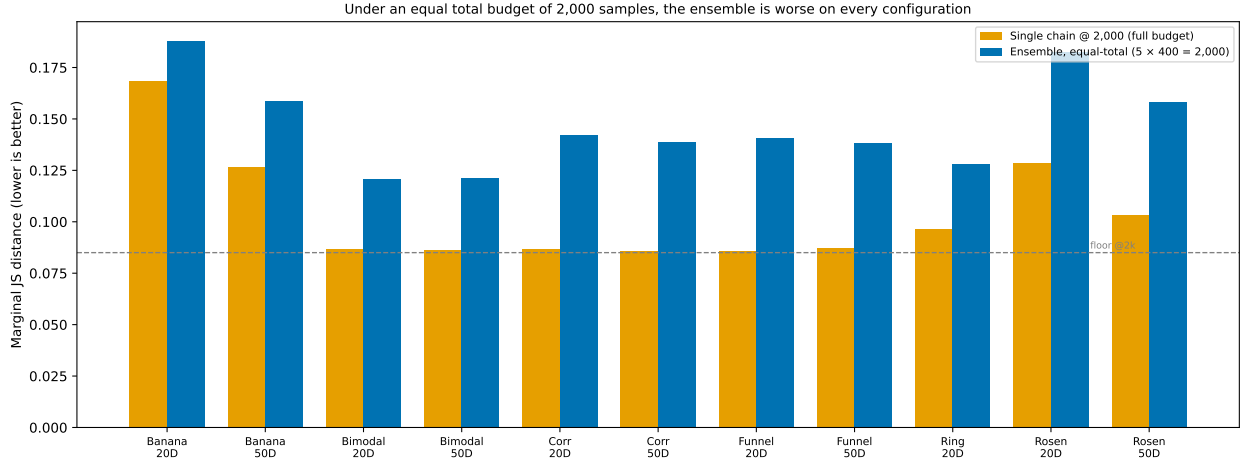


Figure 8: Equal-total budget: with the same 2,000-sample total, the ensemble (five members of 400 samples each) is worse than the single chain on every configuration, by 11% to 64%. The dashed line is the 2,000-sample noise floor.

5.9 Cost

On a single accelerator the equal-per-member ensemble takes about five times the wall-clock time of the single chain, because it runs five separate flowMC instances. The cost of one flowMC run is dominated by its fixed training loops, not by the particle count: on our hardware the 10,000-particle chain runs in about 15 seconds, comparable to the 2,000-particle chain (about 18 seconds; the difference is within run-to-run variation) and roughly six times faster than the ensemble (about 92 seconds). The single chain therefore gets five times the samples at almost no extra cost, while the ensemble pays a fivefold cost for samples that are worse. (With five accelerators the members could run in parallel and the wall-clock would become comparable, but at five times the hardware and still with worse samples.) In none of our settings is the ensemble the better choice. Figure 9 places the three samplers on the accuracy–cost plane.

6 Discussion

6.1 What actually happened

Several findings point to one conclusion, and the controlled experiments now say *why*. The ensemble’s apparent advantage over a small single chain is a sample-count artifact whose size is given in closed form by Equation (2) (Section 5.3); at an equal total budget a single chain is better (Sections 5.2 and 5.8); member diversity does nothing (Section 5.4); and none of the aggregation heuristics helps (Section 5.4). The direct control in Section 5.5 locates the cause — replacing the aggregation with plain uniform pooling returns the accuracy to the single-chain level — and the sampler-free decomposition in Section 5.6 identifies the mechanism. The deficit has two parts. The resampling-with-replacement step duplicates points and thereby doubles the sampler-side variance

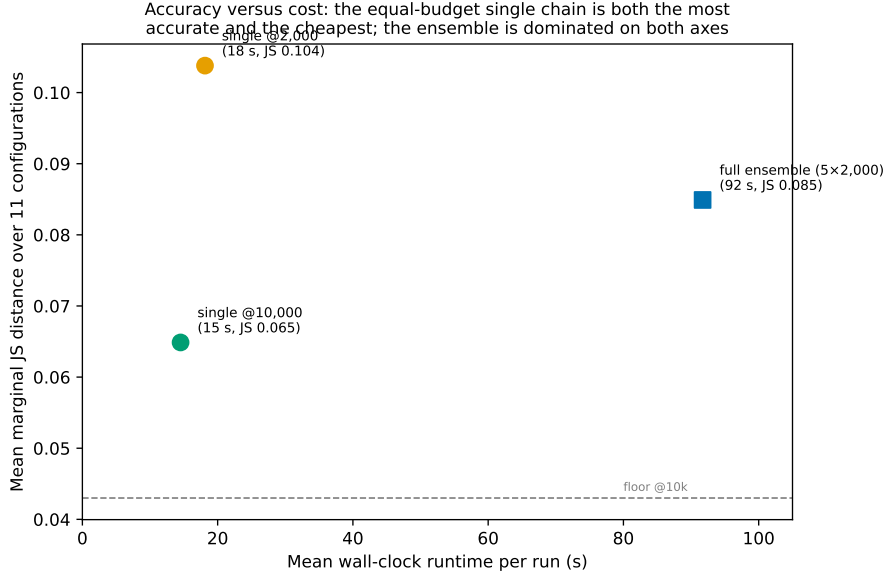


Figure 9: Accuracy versus cost (mean over the eleven configurations; measured wall-clock runtimes). The equal-budget single chain is simultaneously the most accurate and the cheapest option, so the ensemble is dominated on both axes: it costs about six times more and is less accurate.

term of the histogram, an estimator-level loss worth about half of the effective sample, predicted quantitatively by the bootstrap factor in Equation (2) and invisible to an unbiased joint metric. The coverage reweighting (and the tempering schedule, which on healthy members only ever sharpens the target, up to $\pi^{1.23}$) genuinely distorts the sampled law, which the energy distance confirms and which is worst exactly where the cluster structure of the first two coordinates is unbalanced. Applied to exact samples, the two effects reproduce about 90% of the measured ensemble deficit, and with tempering included they reproduce the measured ensemble values on the correlated and funnel targets nearly exactly. The sample-set ESS stays near nominal throughout (9,848 against 9,846 on exact samples), so this diagnostic cannot detect either effect. A uniform pool imposes neither, which is why it matches the single chain under this marginal-JS evaluation.

Looking back, the part that is a little uncomfortable to admit is how convincing the wrong version looked. The first draft of this work reported the 18% improvement with significance on every target and large effect sizes, and all of that was technically correct — it just answered the wrong question. The lesson we take from it is mundane but easy to forget under the pressure of a positive result: before believing a comparison, make the baselines spend the same budget.

6.2 Implication for practice

For the targets and budgets we study, the practical advice is the opposite of the method we set out to build. Given a fixed sampling budget, run one flowMC chain at the full budget instead of splitting or multiplying it across an ensemble. If for some reason the members are pooled, pool them uniformly — concatenation preserves both the law and the effective sample — and do not expect to beat the single chain. Avoid coverage-style reweighting, diagnostic-driven tempering, and in particular any aggregation that resamples with replacement, which by itself costs half of the effective sample under histogram-based metrics. Ensembling in this setting adds compute and complexity and gives nothing back. Three further guidelines concern evaluation rather than sampling. When a histogram-based

divergence such as marginal JS is used to compare samplers, the comparison is only meaningful at matched sample counts, and reporting the finite-sample floor of Equation (2) alongside the results makes the remaining headroom visible at a glance. Pairing the histogram divergence with an unbiased joint metric such as the energy distance separates true distributional error from estimator artifacts, as it did here. And the reference generator itself should be validated against the potential with an exact-draw test (Section 5.7) before any divergence-to-reference number is interpreted as sampler error.

6.3 Scope and limitations

Our headline comparisons concern marginal JS distance on these synthetic targets. We measured the joint-distribution behaviour of the aggregation operators with the energy distance, but only on exact samples (Section 5.6); a joint metric evaluated on the actual flowMC outputs could still reveal mode-coverage effects of ensembling that marginal JS cannot, and remains the natural way to settle whether ensembling helps for genuinely two-dimensional multimodal structure. This caveat is not the same for every target: for the bimodal Gaussian, whose modes are separated along a single axis, the marginal JS on that axis does reflect mode coverage, so our conclusion there is on firmer ground; for the Gaussian ring the marginals are partly washed out, and the audit of Section 5.7 additionally shows that the ring’s released reference differs from its potential, so the single-chain result on the ring should be read as a statement about marginal accuracy against that reference. We used the aggregation mechanisms at fixed default hyperparameters without per-target tuning; tuning might change their contribution, but a mechanism that has to be tuned carefully just to avoid harming the result is of limited practical value, and the exact-sample decomposition shows that two of the three mechanisms (and the shared resampling step) are harmful at these defaults even under ideal conditions. The tempering isolation of Section 5.6 is exact only for the two target families whose tempered laws are available in closed form. Finally, we tested six synthetic targets at 20 and 50 dimensions, one flow family, and ensemble sizes up to twenty members only in the exact-sample setting; real posteriors, other flows, or other aggregation designs may behave differently. Beyond these caveats, the extensions that would sharpen the picture most are evaluating joint metrics on the actual flowMC outputs, repeating the study on real posteriors with funnel-like or multimodal geometry, and re-running the Rosenbrock and ring rows against corrected references.

7 Conclusion

We set out to validate a diversity-enhanced flowMC ensemble, and instead we found that its apparent benefit was an artifact of comparing at unequal sample counts — an artifact whose size is given in closed form by the finite-sample law of the histogram JS estimator, $\mathbb{E}[\widehat{\text{JS}}^2] \approx c(1/N + 1/M)$. With a matched 10,000-sample budget a single flowMC chain matches or beats the full ensemble on every target — winning on nine of eleven configurations and tying on the two banana cases — and lowers the mean marginal JS distance by 24%, while running about six times faster. Taking the ensemble apart shows where the benefit went: diversity does nothing, none of the three aggregation heuristics helps, and replacing the aggregation with plain uniform pooling recovers the single-chain accuracy. Removing flowMC itself then shows why: applied to exact draws, the aggregation operators reproduce about 90% of the deficit, and with the tempering schedule they reproduce the measured ensemble on the correlated and funnel targets. The damage is the sum of a genuine distribution shift from the coverage and tempering reweighting — confirmed by an unbiased energy distance — and an estimator-level loss from resampling with replacement, which halves the effec-

tive histogram sample; neither is visible to the sample-set ESS, and neither is repaired by adding members. An audit of the reference generators adds a final caution: two of the six targets were scored against a law that differs from their potential, and once this is accounted for, only the banana remains genuinely sampler-limited. The recommendation is short: run one flowMC chain at the full budget; if you pool members, pool them uniformly and expect no gain over a single chain; stay away from the resampling-and-reweighting machinery; and when benchmarking with a histogram divergence, match the sample counts, report the floor, validate the reference, and keep an unbiased joint metric alongside. The broad lesson is an old one that still catches us: an ensemble compared against a smaller baseline can look effective for reasons that have nothing to do with the ensemble. The remedy is equally simple: match the budgets, know the finite-sample floor of the metric, and give a plain control the chance to falsify the method early.

Reproducibility

All target definitions, member configurations, per-repetition seeds, the control scripts (single at 10,000, the homogeneous ensemble, and the uniform-pooling control) and the analysis code are available at [13]. Every number reported here can be regenerated from the released seeds. The repository also includes the CSV outputs used to produce Tables 1–3 and the corresponding figures. The sampler-free experiments of Sections 5.3, 5.6 and 5.7 are produced by a self-contained script (`supp_experiments.py`) that ports the released JS estimator, aggregation operators and reference generators to NumPy (the ports are checked for numerical equivalence against the released implementations), together with its CSV outputs (`supp_*.csv`) and the figure-generation code; the table bodies of this paper are generated directly from the CSVs.

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